AgriCien

COMPANY PROFILE 2025

BOOSTING AGRICULTURE PRODUCTIVITY

Agricultura Científica SA Company (AgriCien)

AgriCien company is a global leader in agro-hydrological engineering and regenerative agriculture, delivering transformative solutions for water-challenged landscapes. We specialize in water-cycle restoration, soil regeneration, climate-resilient agricultural systems, and large-scale agro-engineering that addresses the most pressing challenges facing modern agriculture. Our integrated approach combines cutting-edge precision technology with proven regenerative principles to create productive, sustainable agricultural systems that thrive in challenging environments

---

Incorporated in 1992, in San Jose, Costa Rica.

Address: Avenida 31-A, Oficentro Palacio No. 8, La Uruca District, 10107 San Jose City, San Jose Province, Costa Rica.

Strategic Value for Investors

AgriCien's solutions deliver compelling financial and operational advantages that transform agricultural investments into resilient, high-performing assets:

Enhanced Productivity

Produce significantly more biomass with less water through advanced hydrological engineering and soil optimization techniques.

Climate Resilience

Maintain productive capacity during drought periods and climate extremes, ensuring consistent returns regardless of weather variability.

Reduced Operating Costs

Decrease long-term operational expenses and risk exposure through improved water efficiency and reduced input dependency.

Asset Appreciation

Increase fundamental asset value through measurable soil capital appreciation and enhanced land productivity metrics.

Proven Scalability

Scale confidently from 50-hectare pilot projects to 20,000+ hectare estates with consistent, replicable results.

AgriCien

Integrated Group Structure

AgriCien Group operates through three specialized divisions, each delivering critical capabilities that combine to create comprehensive agricultural transformation:

AgriCien Technology

Advanced precision agriculture solutions including Geographic Information Systems (GIS), drone-based monitoring and analysis, GNSS automation for mechanized operations, and real-time data analytics that enable informed decision-making and operational efficiency at scale.

AgriCien Regeneration

Landscape Design Implementation (LDI), comprehensive soil-water restoration programs, holistic planned grazing systems, and biological soil enhancement protocols that rebuild natural capital and ecosystem function for long-term productivity gains.

AgriCien Engineering

Full-spectrum project development including pre-feasibility analysis, hydrological engineering design, master planning and phasing, financial modeling (CAPEX/OPEX/IRR), and construction management for large-scale agricultural infrastructure projects.

AgriCien

Proven Performance Metrics

AgriCien's interventions deliver measurable, transformative improvements across all critical agricultural performance indicators. These results represent typical outcomes from completed projects and demonstrate the substantial value creation potential:

+100%

Water Use Efficiency

From uncontrolled to precision irrigation

2-3x

Biomass Production

From <5 to >9 tonnes grain per hectare

100%

Monitoring, Automation

Gates, valves, flows under remote control

72%

Precision Direct Seeding

Large scale machine automation

Precise water management

Savings in water consumption, energy costs.

AgriCien

LDI Innovation: Transforming Water Economics

Líneas de Infiltración – Infiltration Lines (LDI) represents a paradigm shift in dryland grasslands water management. This engineered approach fundamentally alters landscape hydrology to capture, infiltrate, and store rainfall that would otherwise be lost to runoff and evaporation.

Key Technical Advantages:

- Infiltrates rainfall directly into the soil profile instead of losing it to surface runoff
- Stores additional 1,500–3,000 cubic meters per hectare in the active root zone
- Dramatically reduces drought risk and increases production reliability
- Triples pasture output in dryland and semi-arid environments
- Eliminates or substantially cuts dependence on costly imported fodder

AgriCien

Autonomous Sensor and Actuator Systems

Controlled Environment Agriculture

Advanced autonomous systems integrate real-time sensor networks with precision actuators to monitor and control critical growing parameters in controlled agricultural environments.

These intelligent systems continuously collect environmental data—including temperature, humidity, soil moisture, light intensity, and nutrient levels—enabling automated adjustments that optimize crop growth conditions.

Machine control automation ensures consistent irrigation delivery, climate regulation, and resource management while reducing labor requirements and minimizing human error.

GNSS-Based Precision Land Leveling

Satellite-Guided Precision

GNSS technology enables centimeter-level accuracy for automated soil leveling and earthmoving operations, eliminating field irregularities.

Optimal Water Distribution

Precisely graded fields ensure uniform water distribution across agricultural land, reducing runoff and improving irrigation efficiency by up to 40%.

Enhanced Soil Preparation

Automated land preparation systems optimize soil conditions for planting, ensuring consistent depth and improved crop establishment rates.

AgriCien

Variable-Rate Precision Spraying Systems

Site-Specific Application Technology

Advanced automated spray systems deliver individual products at variable rates based on real-time field mapping and crop requirements, revolutionizing agrochemical application efficiency.

01

Precision Targeting

GPS-guided nozzle control applies chemicals only where needed, reducing waste by 30-50%

02

Individual Product Control

Multiple tanks enable simultaneous management of different agrochemicals with independent dosing

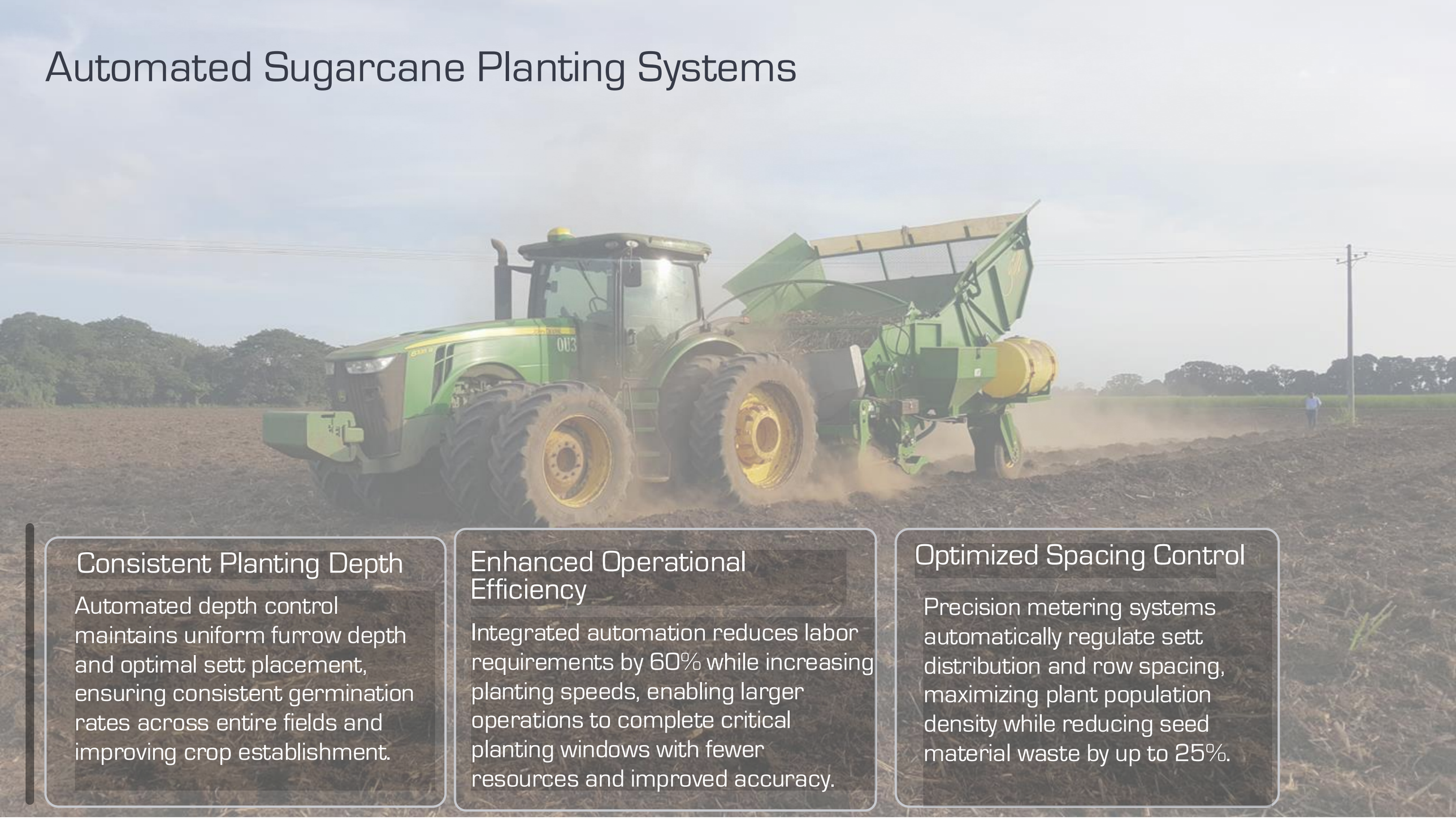
03

Real-Time Adjustment

Sensors detect field conditions and automatically adjust application rates for optimal coverage

AgriCien

Automated Sugarcane Planting Systems



Consistent Planting Depth

Automated depth control maintains uniform furrow depth and optimal sett placement, ensuring consistent germination rates across entire fields and improving crop establishment.

Enhanced Operational Efficiency

Integrated automation reduces labor requirements by 60% while increasing planting speeds, enabling larger operations to complete critical planting windows with fewer resources and improved accuracy.

Optimized Spacing Control

Precision metering systems automatically regulate sett distribution and row spacing, maximizing plant population density while reducing seed material waste by up to 25%.

Proven Performance Metrics

AgriCien's interventions deliver measurable, transformative improvements across all critical agricultural performance indicators. These results represent typical outcomes from completed projects and demonstrate the substantial value creation potential:

10x

Infiltration Rate

From 50 mm/hr to 500+ mm/hr, virtually eliminating runoff and erosion

2-3x

Biomass Production

From 3–7 to 8–15+ tonnes dry matter per hectare annually

4-6x

Carrying Capacity

From 0.6–1.6 to 4–7 animal units per hectare

72% Compaction Reduction

From 1,800–3,000 kPa to under 500 kPa, enabling root penetration, aeration, infiltration

200% Root Depth Increase

From 10–15 cm to 30–50 cm, accessing deeper water reserves and exploring more root zone volume

AgriCien

Strategic Alignment with Policy Priorities

AgriCien's solutions directly address critical national and regional policy objectives, making our projects highly relevant for government partnerships and institutional investment:

Food Security Strategies

Support national food sovereignty goals through increased domestic production capacity and reduced import dependency for critical agricultural commodities.

Water-Efficient Agriculture

Demonstrate viable pathways for productive agriculture in water-stressed zones, unlocking previously marginal lands for food production.

Land Restoration

Support large-scale land restoration commitments and environmental goals while maintaining economic productivity and generating returns.

National Fodder Production

Enable substantial increases in domestic grass production, reducing vulnerability to climate extremes and supply chain disruptions.

Climate Adaptation

Provide concrete climate adaptation solutions that reduce livestock vulnerability and systemic risk in the face of increasing climate variability.

Resilience Frameworks

Integrate with national resilience and disaster prevention frameworks by reducing flood risk, preventing soil erosion, and enhancing landscape water retention.

AgriCien

Our Implementation Process

AgriCien employs a systematic, phased approach that minimizes risk while maximizing learning and adaptation throughout project development:

Pre-Feasibility Assessment

Comprehensive analysis of land characteristics, water availability, soil conditions, climate patterns, and market context to establish project viability and identify constraints.

Feasibility & Engineering

Detailed master planning, hydrological modeling, infrastructure design, and complete financial analysis including CAPEX requirements, OPEX projections, and IRR calculations.

Pilot Implementation

Execution of 50–100-hectare demonstration areas to validate designs, refine techniques, establish baseline metrics and build operational capacity before scaling.

Scaling Operations

Systematic expansion from pilot areas to hundreds or thousands of hectares using proven methodologies, established supply chains, and trained operational teams.

Monitoring & Optimization

Continuous performance monitoring, data collection, adaptive management, and ongoing optimization to maximize returns and ensure long-term sustainability.

AgriCien

Global Track Record

AgriCien has successfully delivered projects across diverse geographies and scales, building a robust portfolio of experience that spans multiple continents and climate zones:

Costa Rica

National LDI implementation program beginning with 45 pilot ranches, achieving documented success that led to nationwide scaling as a model for tropical regenerative agriculture.

Central America

FAO-supported regional adoption across the Dry Corridor, addressing critical food security and climate resilience challenges in one of the region's most vulnerable agricultural zones.

Indonesia

Comprehensive feasibility study for 39,000-hectare large-scale agricultural development, including agronomy, geographical, engineering hydrological analysis, infrastructure planning, and financial modeling.

Abu Dhabi

Design & feasibility services for 230,000-hectare agricultural development land in West Africa, demonstrating capability to support large private-scale agricultural investments in extreme environments.

Latin America, East Africa

Extensive portfolio of farming project development, rehabilitation, mechanization automation, and land-forming engineering across diverse projects ranging from 100 to 10,000s hectares.

AgriCien

Why Choose AgriCien

Proven Results

Documented dryland resilience outcomes with measurable performance improvements across infiltration, biomass, carrying capacity, and soil health metrics.

Integrated Approach

Unique integration of engineering precision with regenerative science, combining the best of both disciplines for comprehensive solutions.

Scalable Solutions

Measurable, low-risk water management solutions that scale from pilot to estate level with consistent, replicable results.

Superior Returns

Higher ROI through increased productivity, reduced operational costs, and lower risk exposure compared to conventional agricultural approaches.

Global Experience

Extensive international experience aligned with government food security priorities, climate adaptation goals, and institutional investment criteria.

Technology-Enabled

Advanced monitoring and transparency through precision agriculture technology, providing investors with real-time visibility into project performance.

AgriCien

Let's Build Resilient Agriculture Together

AgriCien Group is ready to partner with institutional investors, government agencies, and agricultural enterprises to deliver transformative results in water-challenged landscapes worldwide.

Contact our team to discuss your project:

General Inquiries:
proyectos@agricien.com
info@agricien.com

Leadership:
Ing. José María Agüero & Ing. Andrés Agüero
President Founder CEO
jose.aguero@agricien.com. andres.aguero@agricien.com